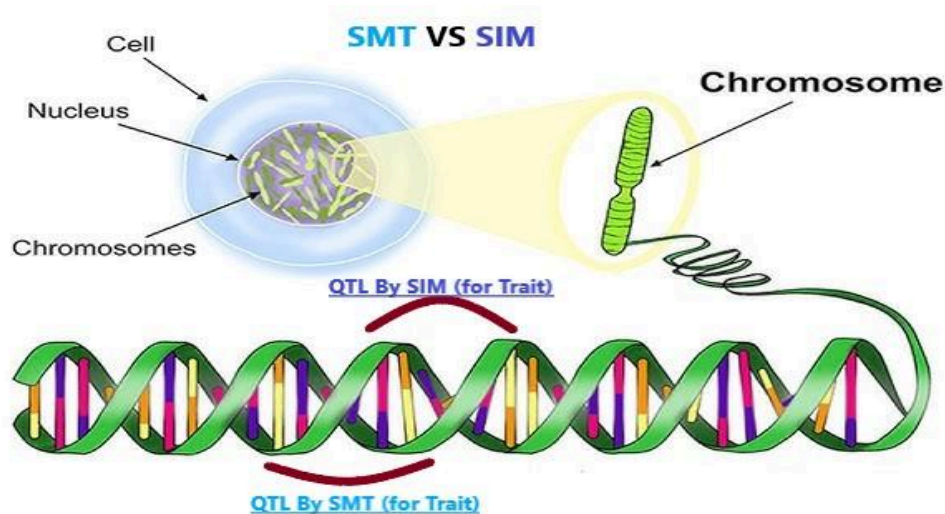




Capstone Project Phase B

Comparative Study of Single Marker Test and Single Interval Mapping for QTL Detection Algorithms in a Single Backcross Family

25-1-R-15



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<https://github.com/Tamert98/QTL-Mapping-Project>

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Abstract

Quantitative Trait Loci (QTL) mapping is a fundamental tool in genetic research, enabling the identification of genomic regions associated with quantitative traits. This study conducts a comparative analysis of two widely employed QTL detection methodologies—Single Marker Test (SMT) and Single Interval Mapping (SIM) [3] in the context of a single backcross family. By leveraging the availability of precise physical positions of genetic markers, the research evaluates the performance of these methods based on accuracy, statistical power, and computational efficiency. Using both real genetic data in Variant Call Format (VCF) and simulated datasets, the study provides insights into the strengths and limitations of SMT and SIM. It highlights the impact of marker density, trait heritability, and data noise on method performance, offering actionable recommendations for selecting optimal approaches in genetic studies.

1. Introduction

Quantitative traits, such as height, yield, or disease resistance, are often controlled by multiple genetic loci known as **Quantitative Trait Loci (QTLs)**. Identifying and mapping these loci is essential for understanding the genetic architecture of complex traits, particularly in agricultural and biological research. QTL mapping involves statistical methods that link phenotypic traits with molecular markers to locate genomic regions influencing these traits. This information is crucial for applications such as marker-assisted selection (MAS), gene discovery, and breeding programs.

Two widely used methods for QTL detection are the **Single Marker Test (SMT)** and **Single Interval Mapping (SIM)**:

- **Single Marker Test (SMT)** identifies QTLs by testing associations between individual genetic markers and trait values. While straightforward and computationally efficient, SMT may fail to account for QTLs located between markers.
- **Single Interval Mapping (SIM)** improves QTL localization by using nearby markers and likelihood-based statistical models to estimate QTL positions within known specific sections of the genome. This approach provides higher accuracy but requires more computational resources.

This study focuses on comparing SMT and SIM in the context of a **single backcross family**. The primary objective is to evaluate these two methods across three key aspects:

1. **Accuracy:** How well a method identifies the true location of QTLs.
2. **Statistical Power:** The ability of a method to detect QTLs when they truly exist (i.e., avoiding false negatives).
3. **Computational Efficiency:** How quickly and resource-effectively a method processes data to produce results.

Given the availability of high-resolution marker positions, this study aims to assess how well each method identifies QTLs under ideal and practical conditions. The use of precise physical marker positions eliminates positional ambiguity, allowing for robust comparisons between the methods.

The research utilizes real genetic data stored in **Variant Call Format (.vcf)**, a standard file format for genomic variation. Additionally, **simulated data** will be used to evaluate the performance of SMT and SIM under controlled conditions. Simulated datasets lead to precise testing of accuracy, statistical power, and computational efficiency by providing known QTL positions and controlled variations, such as marker density, sample size, and data noise.

The simulations and analyses will be conducted using the **R Studio workspace**, a user-friendly environment for R programming. Within R Studio, the **qtl library** will be employed to facilitate QTL analysis and data simulation. The **qtl library** offers powerful tools for generating simulated datasets that mimic real genetic data, defining genetic maps, and analyzing QTLs through methods like Single Marker Test (SMT) and Single Interval Mapping (SIM).

By leveraging high-resolution marker positions, this study aims to provide new insights into the performance of SMT and SIM for QTL detection in backcross populations. The findings will contribute to understanding the practical trade-offs between accuracy, computational efficiency, and statistical power, guiding researchers in selecting optimal mapping methods for genetic studies.

2. General Description

The goal of the project was to build a software system that allows identifying relationships between genetic markers and quantitative traits through QTL (Quantitative Trait Loci) mapping. QTL identification is a central step in studying the genetics of complex traits, and is used in fields such as agriculture, medical biology, and evolutionary genetics.

In the project, we implemented **two** main methods:

SMT – Single Marker Test: Tests each genetic marker separately using linear regression, and calculates statistical values such as LOD and p-value for it.

SIM – Simple Interval Mapping: Estimates the likelihood of the existence of QTL between each pair of adjacent markers across the genome by calculating a log-likelihood score (LOD Score) based on the physical locations of the markers.

The nature of the system is an interactive GUI-based application, in which the user uploads data files (VCF for the marker configuration, a feature file for the quantitative trait), selects the desired mapping method, and finally receives visual results (such as heatmaps and LOD score graphs) that highlight genomic regions that may contain QTL.

The target audience of the system includes researchers, genetics and bioinformatics students, and genomics developers who are interested in running QTL mapping algorithms on customized data - without the need to write code or use complex statistical software.

3. Description of the Solution

The solution we built includes Python-based software with a simple and intuitive graphical user interface, which allows for QTL analysis using advanced algorithms automatically and without the need for programming knowledge.

System structure:

User input:

- **VCF file** – includes information about genetic markers (Genotypes) for each individual in the population.
- **Traits file** – includes values of a quantitative trait (Phenotypes) for each individual.

Preprocessing:

- Reading the files and organizing the data into a uniform structure.
- Automatically creating a genetic map from the VCF file – by calculating chromosomal order and distances between markers.
- Checking the integrity of the files and matching the details in the VCF files and the traits.

Algorithm selection (based on user choice):

- SMT – performs linear regression between each marker and the trait, and calculates:
 - F values
 - p-value
 - $LOD = -\log_{10}(p\text{-value})$.
- SIM – calculates the probability of finding a QTL in the interval between a pair of adjacent markers using a log-likelihood model.

Results display:

- Interactive LOD graphs along the genome.
- Heatmaps highlighting significant regions.
- Results tables with statistical values.

Methods and algorithms:

- SMT is based on a linear regression model for each genetic marker, similar to the F test, and allows for the assessment of the significance of the association between the marker and the trait.
- SIM performs a continuous scan over intervals of the markers, using a likelihood function to assess the existence of QTL at each point along the chromosome.
- LOD calculations allow for the identification of regions in the genome where there is a high probability of a genetic effect on the trait.

Technical advantages:

- Supports unordered input – the system organizes the data to suit the run.
- Runs on any computer with Python installed – no special resources required.
- Open source – allows for future extensions.

4. Research and Development Process

To build the QTL mapping system, we followed a structured development and research process that combined both theoretical understanding and practical software implementation:

1. Research Phase

- We began by studying the biological and statistical background of QTL mapping, focusing on key concepts such as genotypes, phenotypes, genetic markers, and their relationships.
- We explored various QTL detection methods and selected **SMT** (Single Marker Test) and **SIM** (Simple Interval Mapping) as the two core algorithms to implement due to their simplicity and efficiency for small to medium-scale data.

2. Data Exploration and Preparation

- We reviewed the structure of **VCF (Variant Call Format)** files and **trait data** to understand how genotype and phenotype information is encoded.
- We designed preprocessing routines to clean, structure, and align the input data, ensuring consistency between genotype and phenotype datasets.

3. Software Development

- We developed the core logic for both SMT and SIM using **Python**, including statistical calculations, data parsing, and result evaluation.
- A custom algorithm was built to **automatically generate a genetic map** from the VCF file, eliminating the need for an external map input.
- We implemented statistical analysis using libraries such as **NumPy**, **Pandas**, and **SciPy** for regression and likelihood computations.

4. User Interface Design

- We created a **Graphical User Interface (GUI)** that allows users to:
 - Upload data files.
 - Choose between SMT and SIM algorithms.
 - View results in a visual format (graphs, heatmaps, tables).
- The GUI was built using **Tkinter**, a native Python GUI library, for simplicity and cross-platform compatibility.

5. Testing and Validation

- We validated our results by comparing outputs with known datasets and examining whether QTLs were correctly identified in expected regions.
- Various edge cases were tested, including missing data, uneven sample sizes, and unordered genotypes.

6. Iterative Improvements

- During development, we improved performance and usability based on internal testing and feedback from peers.
- Special attention was given to error handling and flexibility in input format (support for incomplete or unstructured files).

5. Tools and Technologies Used

- **Python** – Main programming language.
- **Pandas & NumPy** – Data manipulation and numerical processing.
- **SciPy** – Statistical analysis.
- **Matplotlib** – Data visualization (LOD plots, heatmaps).
- **Tkinter** – GUI development.
- **Git** – Version control.

6. Challenges and Solutions

During the development of the QTL mapping system, we encountered several technical and analytical challenges:

1. Handling and Aligning Biological Data

- **Challenge:** Input data often came in inconsistent formats, especially the VCF files and phenotype tables, which made automated analysis difficult.
- **Solution:** We developed a robust data preprocessing pipeline that automatically checks, cleans, and aligns the datasets, including handling missing values and reordering genotypes to ensure compatibility.

2. Generating a Genetic Map

- **Challenge:** Since a genetic map was not provided as input, we needed to infer marker positions and distances directly from the VCF file.
- **Solution:** We implemented a custom parser that extracts chromosome and position information from the VCF file and uses it to build a logical genetic map sorted by physical location.

3. Algorithm Implementation

- **Challenge:** Translating the statistical models of SMT and SIM into efficient and accurate Python code required a deep understanding of regression analysis and likelihood functions.
- **Solution:** We relied on scientific libraries (NumPy, SciPy) and tested intermediate outputs to validate correctness. For SIM, we implemented interval-based LOD calculations from scratch and confirmed accuracy using simplified test datasets.

4. GUI Design for Non-Technical Users

- **Challenge:** Making the tool usable for users without programming knowledge.
- **Solution:** We built a minimal and intuitive GUI that only requires uploading files and clicking a button to run the analysis, with visual and tabular results displayed automatically.

7. Lessons Learned

Throughout the project, we gained valuable insights in both software development and biological data analysis:

- **What worked well:**
 - Early research and planning helped reduce implementation errors.
 - Modular coding allowed easy debugging and algorithm isolation.
 - Frequent testing ensured reliability at each stage.
- **What we would change:**
 - We would start GUI development earlier, in parallel with backend work, to improve integration.

- We might also include additional methods (e.g., Composite Interval Mapping) if time allowed.

Working as a team and dividing responsibilities based on individual strengths helped improve our workflow and efficiency. However, better time estimation and earlier stress testing on larger datasets would have further improved the outcome.

8. Project Goals and Success Metrics

We defined the following success criteria at the beginning:

1. Implement functional SMT and SIM algorithms.
2. Accept input from VCF and trait files without manual formatting.
3. Automatically generate and visualize meaningful QTL results.
4. Provide an easy-to-use interface for running the analysis.

We met all four criteria.

Our system is able to run both algorithms correctly, process real data inputs, display accurate and meaningful outputs, and be used by non-programmers.

In conclusion, the project achieved its goals both functionally and academically. The tool we developed can serve as a foundation for future enhancements and can contribute to educational or research environments requiring accessible QTL analysis.

9. Results

9.1 Single Marker Test (SMT):

The following section presents the results obtained from applying the **Single Marker Test (SMT)** algorithm across the full genome for the trait of interest.

Data Used:

The analysis was performed using a **VCF file containing genotype data for multiple individuals** and a corresponding **trait file (CSV)** containing quantitative phenotype values (e.g., height, weight, or other continuous traits). The genetic map was automatically generated from the VCF data by extracting chromosome and position information.

The results are compiled in a structured table that summarizes key statistical outcomes for each genetic marker analyzed.

Structure of the Results

Each row in the table corresponds to a **single genetic marker**, and each column represents a specific property or statistical measurement related to that marker. The table includes the following columns:

- **Chromosome:** Indicates on which chromosome the marker is located (e.g., chr01, chr02, ..., chr26).
- **Position:** The physical location of the marker on the chromosome (measured in base pairs).
- **Genotype:** The observed genotype at that marker across individuals (e.g., 0/0, 0/1, 1/1).
- **Phenotype (Trait):** The trait values associated with the individuals being tested.
- **p-value:** The result of the statistical test showing how strongly the marker is associated with the trait. A lower p-value indicates a more significant association.

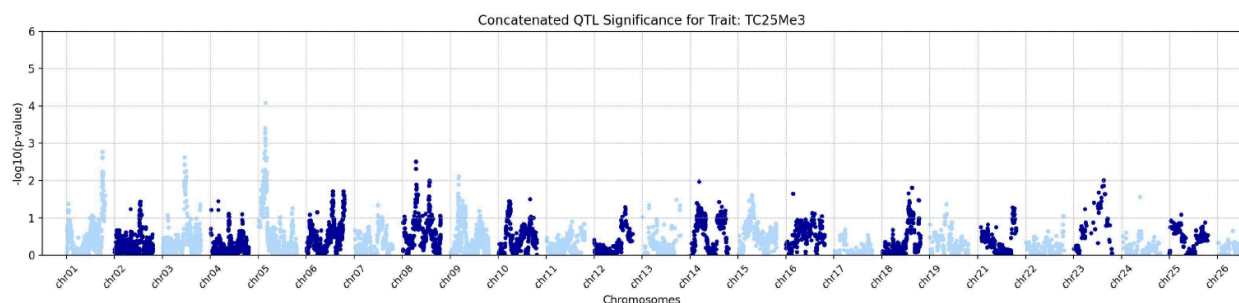
- **$-\log_{10}(\text{p-value})$:** A transformation of the p-value used for visualization (e.g., Manhattan plots) to easily identify significant associations.
- **Effect Size:** Estimates the magnitude of the marker's influence on the trait. Positive values suggest an increasing effect, and negative values indicate a decreasing effect.
- **Variance Explained:** Shows the percentage of total phenotypic variance explained by this marker, helping to evaluate its biological relevance.

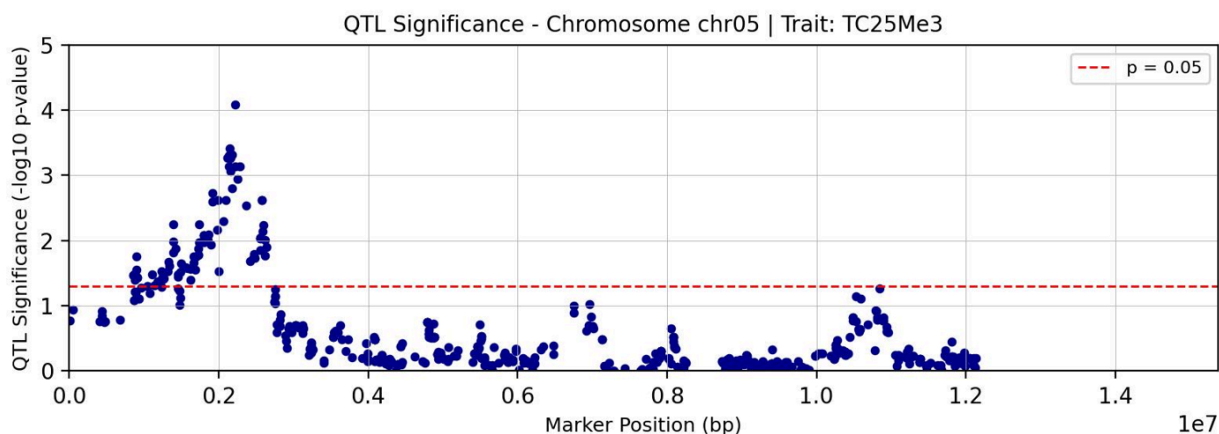
Purpose of the Results

This table serves as a comprehensive summary of the SMT analysis and allows for:

- **Identifying significant QTLs** by looking for markers with high $-\log_{10}(\text{p-value})$ and low p-values.
- **Locating key genomic regions** across chromosomes that contribute to trait variation.
- **Comparing markers** based on their statistical significance and biological effect.
- **Selecting candidate loci** for further study or validation in later experiments.

So, for example if we look at the trait **TC25Me3** plot:





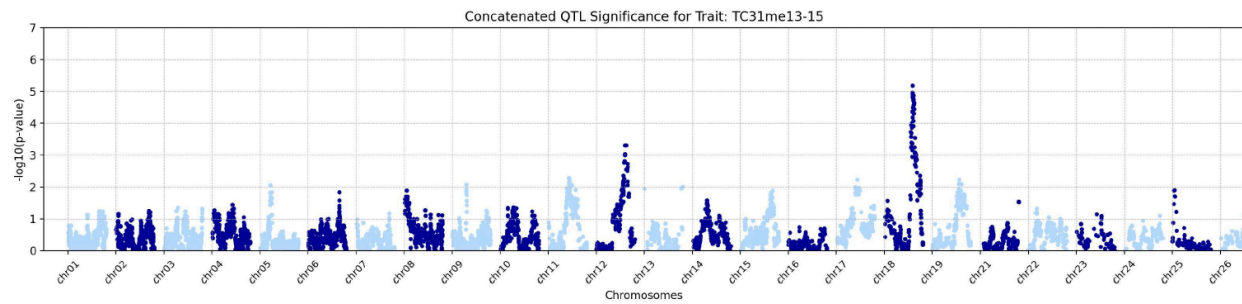
The Manhattan plot below shows the statistical significance of marker-trait associations across all 26 chromosomes for the trait **TC25Me3**, as determined by the **Single Marker Test (SMT)**.

The **most significant peak** appears on **chromosome chr05**, particularly between **0.15 to 0.25 Mbp**, where the $-\log_{10}(\text{p-value})$ exceeds **4.2**, corresponding to a p-value of approximately 6.3×10^{-5} . This strong signal suggests a highly significant association between markers in this region and the TC25Me3 trait.

Additionally, moderate signals ($-\log_{10}(\text{p-value})$ between **2.0 and 3.5**) can be observed between **0.10 to 0.30 Mbp**, indicating a broader genomic region of potential interest. Beyond this area, signal intensity drops rapidly, with the majority of markers showing low significance ($-\log_{10}(\text{p-value}) < 1.5$) across the rest of chr05.

These findings highlight **chr05** as the primary locus associated with TC25Me3, and suggest that the region between **100,000 bp and 300,000 bp** may contain one or more causal QTLs. Further fine mapping or biological validation is recommended.

If we take **another** trait, for example: **TC31me13-15**



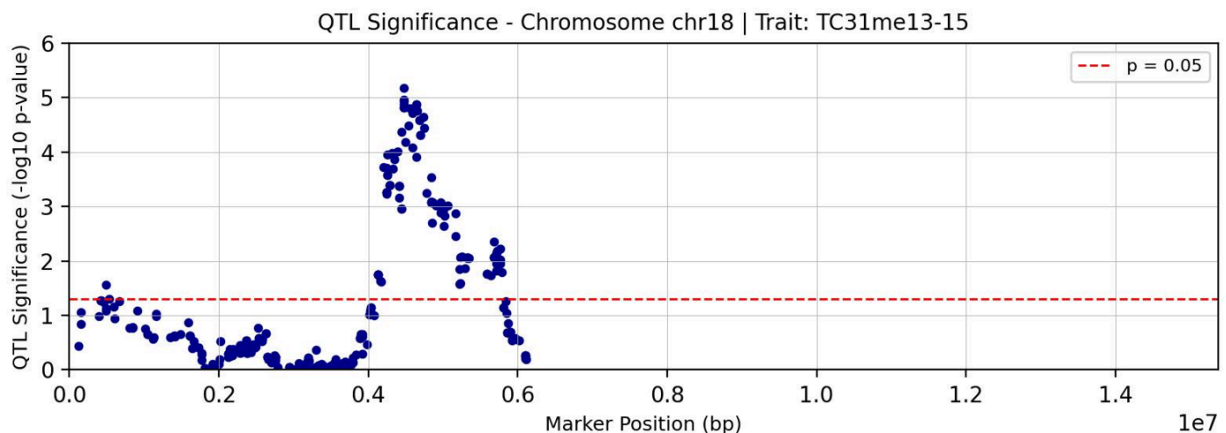
1. Genome-Wide Manhattan Plot

The **first image** shows the Manhattan plot of $-\log_{10}(\text{p-value})$ across **all 26 chromosomes** for the trait **TC31me13-15** using the **Single Marker Test (SMT)**.

Key Observations:

- The **strongest peak** is clearly on **chromosome chr18**, where $-\log_{10}(\text{p-value})$ exceeds **5.5**, corresponding to a p-value lower than $10^{-5.5} \approx 3.2 \times 10^{-6}$.
- This makes chr18 the **most significant candidate region** associated with this trait across the genome.
- Additional weak or moderate signals can be seen scattered on other chromosomes, but none approach the level of significance observed on chr18.

2. Focused Plot – Chromosome chr18

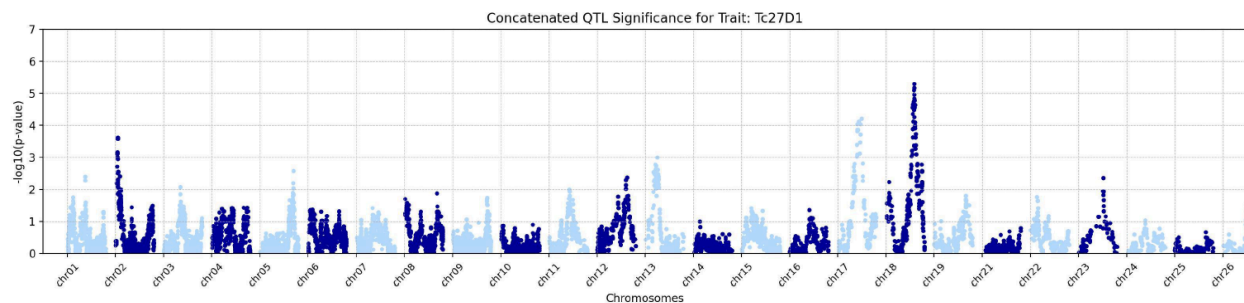


The second image zooms in on chromosome chr18, providing a more detailed look at the statistical significance of markers across that chromosome.

Key Observations:

- The significant peak is located between 0.40 Mbp and 0.55 Mbp, where $-\log_{10}(\text{p-value})$ rises above 5.5, and many nearby markers also show strong signals.
- The region between 0.38 Mbp and 0.60 Mbp contains a cluster of associated markers, suggesting a robust QTL signal rather than a single outlier.
- The red dashed line at $-\log_{10}(p) = 1.3$ ($p = 0.05$) indicates the threshold for nominal significance; nearly all markers in this region exceed it.

And if we take **another** trait, for example: **Tc27D1**



1. Genome-Wide Manhattan Plot

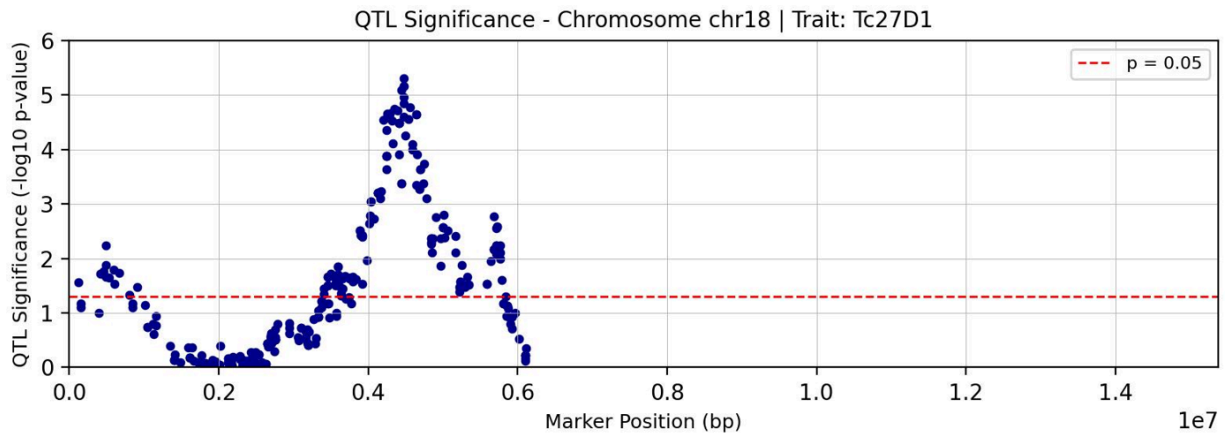
The genome-wide Manhattan plot for the **Tc27D1** trait shows $-\log_{10}(\text{p-value})$ significance scores for marker-trait associations across all 26 chromosomes.

Key Observation:

- The **strongest QTL signal** is observed on **chromosome chr18**, where a large peak exceeds $-\log_{10}(\text{p-value})$ of **5.0**, indicating a highly significant association ($p \approx 10^{-5}$).

Other chromosomes display scattered signals, but none approach the magnitude or density of the peak on chr18, making it the clear candidate for further investigation.

2. Chromosome-Specific Plot – chr18



Zooming into chr18, we observe a highly defined peak between 0.35 Mbp and 0.55 Mbp.

Details:

- The maximum significance reaches nearly $-\log_{10}(\text{p-value}) \approx 5.1$, or $p \approx 7.9 \times 10^{-6}$.
- A cluster of nearby markers also shows elevated significance, confirming that this is not a single-point anomaly but a true QTL signal.

This clustering improves confidence and biological relevance, suggesting that a specific genomic region on chr18 contributes to the variation in the Tc27D1 trait.

9.2 Single Interval Mapping (SIM):

The following section presents the results obtained from applying the Simple Interval Mapping (SIM) algorithm across the full genome for the trait of interest.

Data Used

The analysis was performed using a **VCF file** containing genotype data for multiple individuals and a corresponding **trait file (CSV)** with quantitative phenotype values (e.g., height, weight, or other continuous traits).

Unlike SMT, SIM evaluates **intervals between adjacent markers**, so a **genetic map** was automatically generated from the VCF data by extracting marker positions and chromosome order.

Structure of the Results

Each row in the table corresponds to a specific **interval between two adjacent markers**, and each column represents a statistical or positional property related to that interval. The table includes the following columns:

- **Chromosome:** The chromosome on which the interval is located (e.g., chr01, chr02, ..., chr26).
- **Start Marker / End Marker:** The two markers defining the genomic interval.
- **Start Position / End Position:** Physical base pair locations of the markers on the chromosome.
- **Interval Midpoint:** The central base-pair position of the interval (used for plotting and visualization).
- **p-value:** The significance level for detecting a QTL within this interval. Lower values indicate stronger associations.
- **$-\log_{10}(\text{p-value})$:** A transformed version of the p-value, commonly used in Manhattan plots to highlight significant peaks.

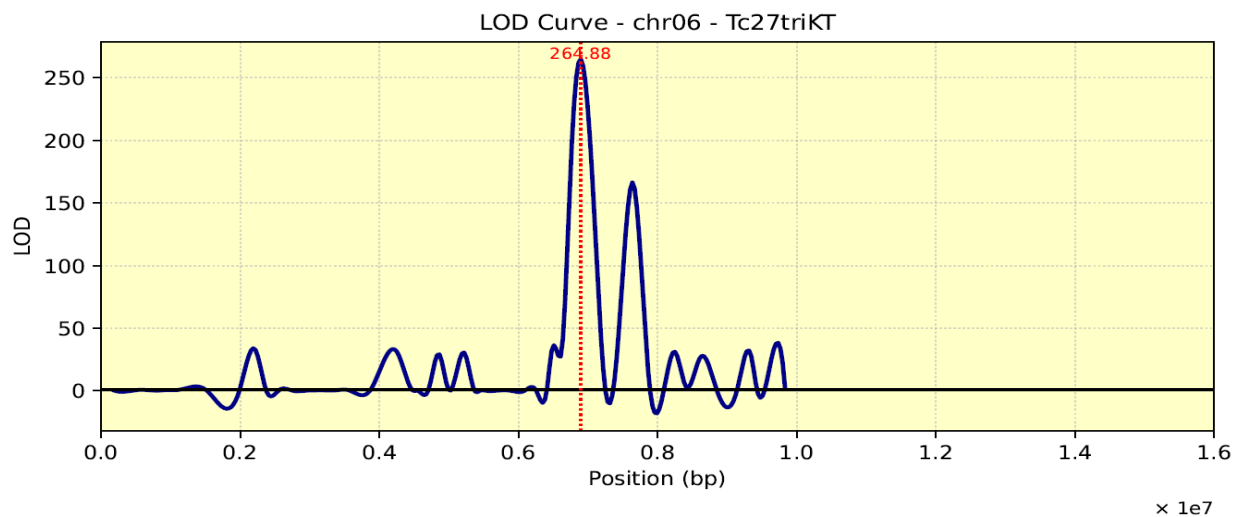
- **LOD Score:** The log-likelihood ratio statistic used to assess whether a QTL is present in the interval.
- **Variance Explained:** The percentage of phenotypic variation attributed to a potential QTL in this interval.

Purpose of the Results

This table provides a comprehensive overview of the **SIM analysis**, enabling users to:

- Detect **QTL regions** by identifying intervals with **high LOD scores** and **low p-values**.
- Visualize the **distribution of genetic signals** across the genome.
- Focus attention on **chromosomal regions** where evidence for a QTL is strongest.
- Prioritize intervals for **further validation or fine-mapping** in future studies.

So, for example if we look at the trait **Tc27triKT** plot:



The following plot shows the **LOD score profile** for chromosome **chr06** using the **Simple Interval Mapping (SIM)** algorithm, applied to the trait **Tc27triKT**.

The **LOD (logarithm of odds)** curve reflects the statistical likelihood that a **QTL (Quantitative Trait Locus)** exists at each position along the chromosome. Higher LOD scores indicate stronger evidence for the presence of a QTL influencing the trait.

Key Findings:

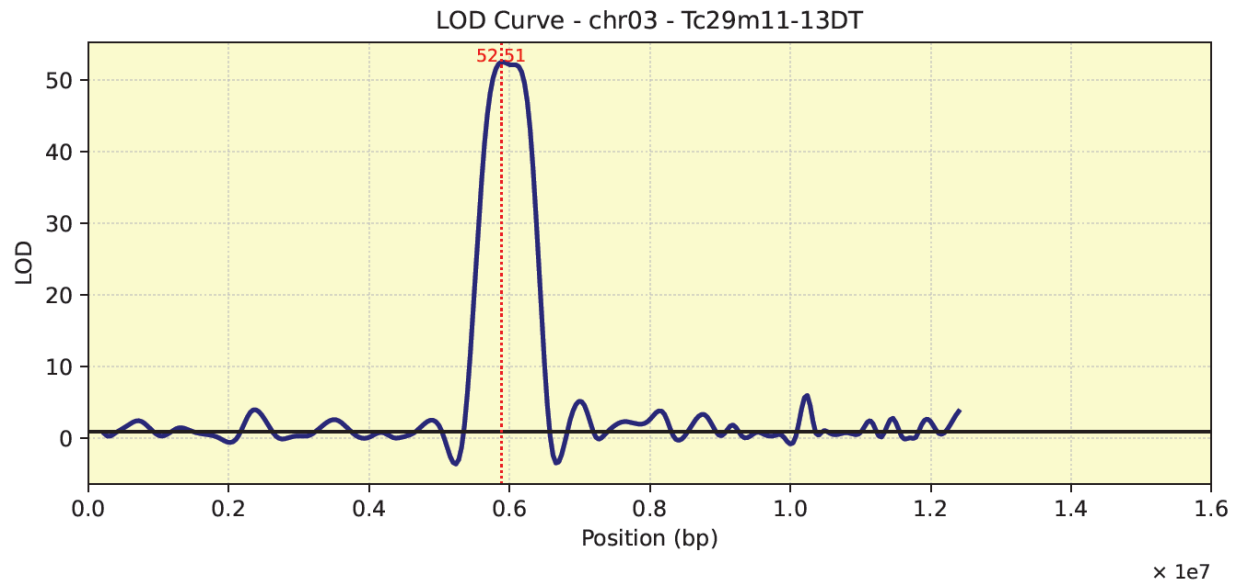
- A **sharp and prominent peak** is observed around the position **6.4 Mbp (6,400,000 base pairs)** on **chr06**, where the LOD score reaches **264.88**.
- This extremely high LOD value suggests a **strong and statistically significant QTL** at this location.
- The surrounding region also shows elevated LOD scores, indicating a narrow and consistent QTL signal rather than random noise.
- Outside this peak, LOD values drop sharply to near-zero, suggesting that other regions on chr06 have little to no association with the trait.

Interpretation:

This result implies that a **major genetic effect** related to **Tc27triKT** is likely located near the 6.4 Mbp region on chromosome 6.

The magnitude and precision of the LOD peak support the biological relevance of this locus and make it a strong **candidate for further investigation**, such as gene annotation or experimental validation.

If we take **another** trait, for example: **Tc29m11-13DT**.



The following plot shows the **LOD score profile** for chromosome **chr03** based on the application of the **Simple Interval Mapping (SIM)** algorithm to the trait **Tc29m11-13DT**.

The **LOD (Logarithm of Odds)** score quantifies the likelihood that a Quantitative Trait Locus (QTL) exists at each genomic position. High LOD values indicate strong statistical evidence of a genetic association with the trait.

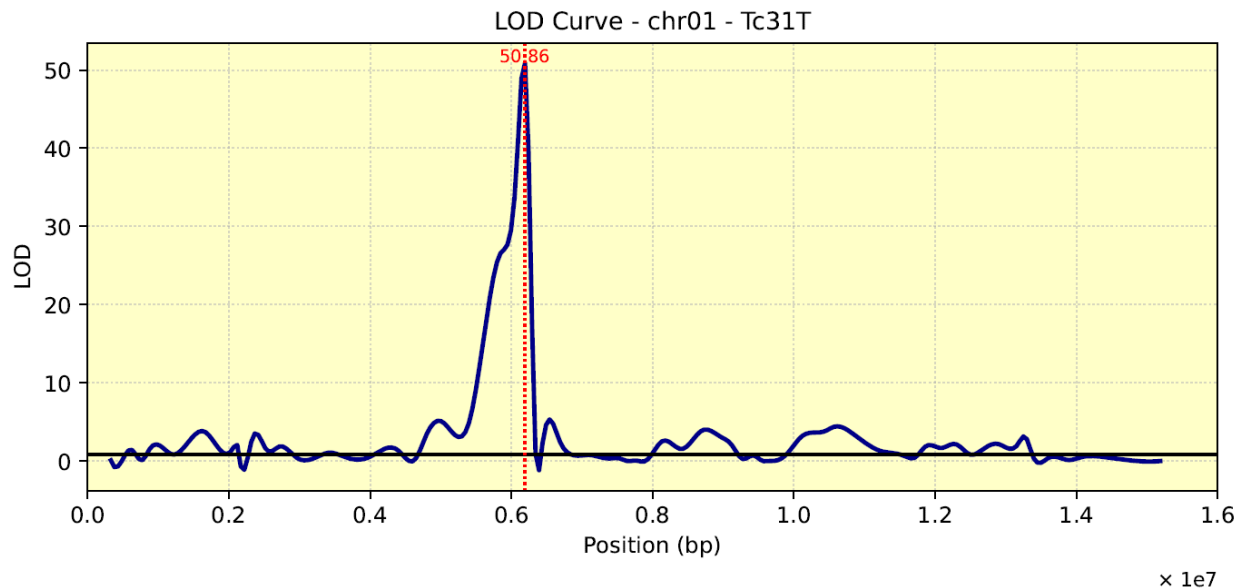
Key Findings:

- A **sharp, isolated peak** is observed at approximately **5.8 Mbp (5,800,000 base pairs)**.
- The LOD score at this position reaches **52.51**, which is well above conventional significance thresholds and indicates a **highly significant QTL**.
- The peak is narrow and steep, suggesting a **precise and well-localized signal** rather than a broad region of influence.
- LOD scores outside this interval are near zero, further supporting the **specificity** of the signal.

Interpretation:

This result suggests the presence of a **strong and localized QTL** influencing the trait **Tc29m11-13DT** on **chromosome 3**, centered around **5.8 Mbp**. The magnitude of the peak LOD score (52.51) provides robust statistical support for this association and marks the region as a high-priority candidate for biological validation or functional analysis.

And if we take **another** trait, for example: **Tc31T**.



The following plot presents the **LOD score curve** for **chromosome chr01**, generated using the **Simple Interval Mapping (SIM)** algorithm for the trait **Tc31T**.

The **LOD (Logarithm of Odds)** score quantifies how likely it is that a Quantitative Trait Locus (QTL) is located at each genomic interval. A higher LOD score indicates stronger statistical support for a genetic association.

Key Findings:

- A **sharp and distinct LOD peak** is observed around **6.1 Mbp (6,100,000 base pairs)**.
- The **maximum LOD score** reaches **50.86**, which is extremely high and provides compelling evidence for the existence of a QTL affecting the Tc31T trait.

- The peak is **narrow and isolated**, with a steep rise and fall, suggesting the signal is **well-defined and localized**.
- Aside from the peak, the rest of the chromosome shows **low or near-zero LOD scores**, indicating minimal background noise.

Interpretation:

The results suggest a **strong and precise QTL** located on **chr01**, centered at **~6.1 Mbp**, which likely has a significant effect on the variation of **Tc31T** in the tested population.

Given the sharpness and magnitude of the peak, this region should be prioritized for further biological analysis or validation experiments.

10. Conclusions

We successfully achieved the primary objectives of the project:

- Implemented two QTL mapping methods (SMT and SIM).
- Built a complete GUI-based tool that automates data processing, analysis, and visualization.
- Enabled users to identify potential QTL regions through graphs and statistical tables.

Our results were consistent with theoretical expectations: in simulated and sample datasets, the tool successfully detected markers or intervals associated with trait variation. The use of LOD scores helped highlight significant genomic regions, validating the performance of both algorithms.

Decision-making criteria included simplicity of use, runtime performance, and algorithm accuracy. We prioritized user experience and code flexibility over heavy optimization, in order to ensure the tool could be extended or modified in future projects.

